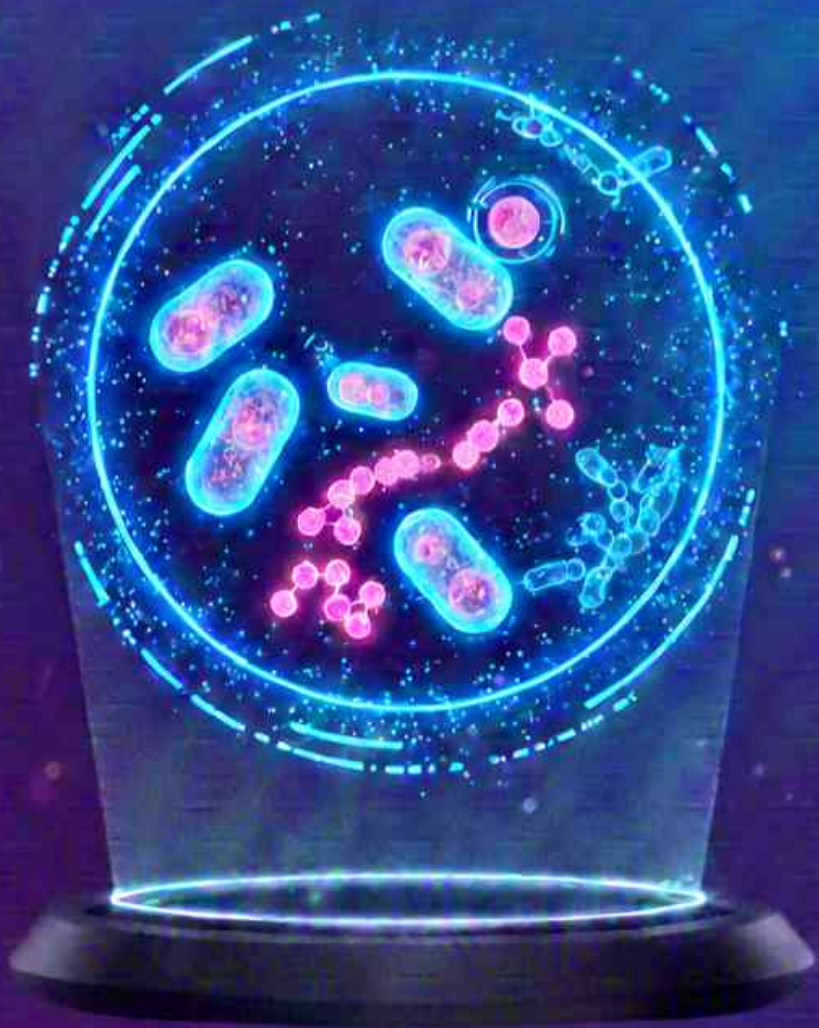




# PRACTICAL SECTIONS

Microbial Physiology (Bacteria)



## → Microbial Physiology ←

### Section 1:

#### → Essential Nutritional Requirements for Microbial Growth

Microorganisms require several essential substances from their environment to support growth, metabolism, and reproduction.

These include:-

#### I) Carbon Source

- Carbon is the fundamental building block of all organic molecules.
- Microorganisms can obtain carbon from different sources:
  - Organic sources: Ex Glucose - Sucrose - Maltos.
  - Inorganic sources: Ex Carbon dioxide ( $\text{CO}_2$ ).
- Based on their carbon requirements, bacteria are classified into:
  - Heterotrophic bacteria: require complex organic carbon compounds.
  - Autotrophic bacteria: Utilize inorganic carbon (Ex  $\text{CO}_2$ ) as their carbon source.

#### II) Nitrogen Source

- Nitrogen is essential for the synthesis of proteins - nucleic acids - and other cellular components.
- Microorganisms can use different nitrogen sources, such as ammonia - nitrates - or organic nitrogenous compounds.

#### III) Non-Metallic Elements

- These are required in smaller amounts but are vital for cellular processes.  
Ex Sulfur - Phosphorus.



#### IV) Metallic Elements.

• These act as Cofactors in enzyme systems and help maintain Osmotic balance.

Ex. Calcium ( $\text{Ca}^{2+}$ ) - Sodium ( $\text{Na}^+$ ) - Potassium ( $\text{K}^+$ ).

#### V) Vitamin Source.

• Vitamins serve as Coenzymes or enzyme cofactors, assisting in various metabolic reactions.

• Many microorganisms require external Vitamin Sources for proper growth.

#### VI) Energy Source.

• Energy is needed to drive biosynthetic and metabolic processes.

• Microorganisms can obtain energy in two major ways:

→ Phototrophs: Derive energy from light through Photosynthesis Ex. Cyanobacteria.

→ Chemotrophs: Derive energy from chemical reactions (Oxidation of organic or inorganic compounds).

#### VII) Water

• Water is the universal solvent and the medium in which all biochemical reactions occur.

• It is indispensable for microbial life.

## → Environmental Conditions Required for Microbial Growth.

### I) PH

- Bacteria: optimum pH 7-7.5.
- Fungi: optimum pH 5-6.5.

### II) Temperature

- Bacteria: optimum temperature  $37^{\circ}\text{C}$ .
- Fungi: optimum temperature  $25-28^{\circ}\text{C}$ .

### III) Incubation Period

- Bacteria: 24-48 hours.
- Fungi: 3-7 days.

Note These ranges vary depending on the species and environmental adaptations.

## → Culture media

Culture media are nutrient mixtures prepared in the laboratory to support the growth and multiplication of microorganisms.

• They are classified into several types:

### I) Routine (General-Purpose) Media

- Contain basic nutrients required for microbial growth.  
Ex. Nutrient Agar.



## II) Selective media.

- Contain specific chemicals or environmental conditions that favor the growth of certain microorganisms while inhibiting other.

Ex.

→ MacConkey Agar:

- Selective for Gram-negative bacteria. (+)
- Contains crystal ~~violet~~ violet which inhibits Gram-positive bacteria. (-)

→ Mannitol Salt Agar (MSA):

- Contains 7.5% NaCl.
- Selective for Gram-positive bacteria (tolerant to high salt) (+)
- Inhibits most Gram-negative bacteria. (-)

## III) Differential Media.

- Allow the growth of multiple types of microorganisms but differentiate them based on biochemical properties:  
(Ex. <sup>biochemical</sup> Colony Color - medium change.)

Example.

→ MacConkey Agar: [selective + Differential].

- Contains Lactose as a fermentable sugar.
- Contains Neutral Red as a pH indicator.
- Lactose fermenters → produce acid +  $H_2O$  → Colonies turn pink/red.
- Non-lactose fermenters → Colonies appear colorless/cream.

#### IV) Blood Agar Medium.

- \* Prepared by mixing nutrient agar with 5% sterile animal blood.
  - \* It is both enriched (supports fastidious organisms) and differential (based on hemolysis).
  - \* Differentiates bacteria according to their ability to lyse red blood cells (RBCs):
    - $\beta$ -hemolysis [Complete hemolysis]:
      - Complete destruction of RBCs and hemoglobin.
      - Produces a clear-transparent zone around colonies.
    - $\alpha$ -hemolysis: (Partial hemolysis)
      - Partial destruction of hemoglobin.
      - Produces a greenish zone around colonies.
    - $\gamma$ -hemolysis:
      - No change in RBCs.
      - The medium remains unchanged around the colonies.
- 



## Section 2

### → Physical Factors Affecting Microbial Growth.

#### I) Temperature:

- Growth depends on enzyme activity affected by temp.
- High temp → enzyme denaturation → Cell death
- Low temp → enzyme inactivation → slow metabolism.
- Bacteria grow between  $-5^{\circ}\text{C}$  to  $80^{\circ}\text{C}$ , each with its own optimum.

#### Types:

##### → Psychrophilic:

- Grow  $-5^{\circ}$  to  $20^{\circ}\text{C}$  - optimum  $0^{\circ}$  to  $5^{\circ}\text{C}$ .

##### → Mesophilic:

- Grow  $20$  to  $45^{\circ}\text{C}$  - optimum  $37^{\circ}\text{C}$ .

##### → Thermophilic:

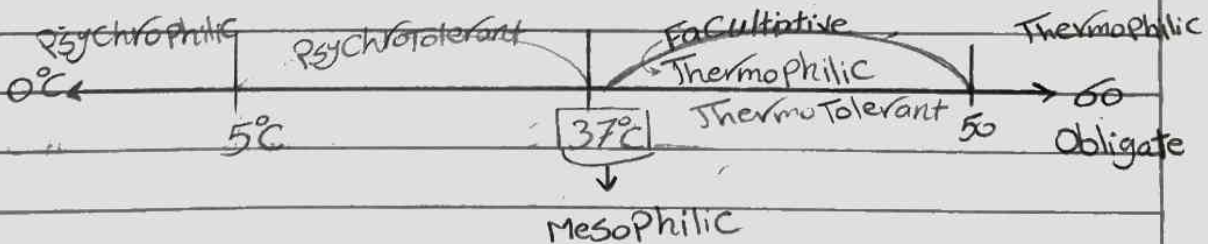
- Grow  $> 35^{\circ}\text{C}$

##### → Facultative:

- Grow  $45$  to  $60^{\circ}\text{C}$  [tolerant]
- optimum  $37^{\circ}\text{C}$ .

##### → Obligate:

- Grow  $50$  to  $60^{\circ}\text{C}$



## II) pH

- Growth and Survival depend on environmental pH.
- Each species grows best in a specific range.

General Ranges: (6.5 → 9)

- Bacteria → pH (4 → 9) - optimum (6.5 - 7.5) ⇒ 7 pH

- Fungi & yeasts → pH (4 - 5.5) (acidic environment)

→ Bacteria are divided according to their growth at different pH levels into:

→ Acidophilic bacteria:

- Grow only at low pH levels [acidic medium]

→ Alkalophilic bacteria:

- Grow only at high pH levels [alkaline medium]

→ Neutrophilic bacteria:

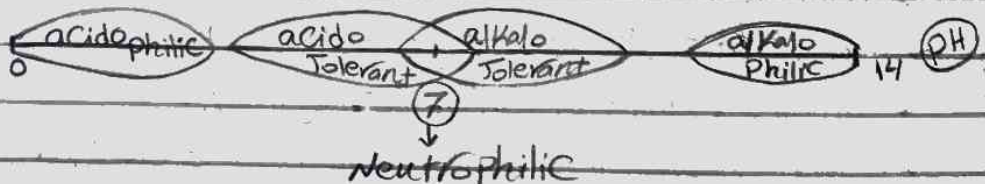
- Grow only at intermediate pH levels [neutral medium]

→ Acidotolerant bacteria:

- Grow at intermediate pH levels and tolerant acidic medium.

→ Alkalotolerant bacteria:

- Grow at intermediate pH levels and tolerant Alkaline medium.





### III) Oxygen Requirements.

- Microorganisms vary in their ability to use free oxygen ( $O_2$ ) for cellular respiration.

→ This variation is due to differences in their biooxidative enzyme systems.

→ According to their  $O_2$  requirements, microorganisms are divided into five major groups:

#### → Aerobic bacteria :

- Need atmospheric oxygen for growth.
- Use  $O_2$  as the final electron acceptor in oxidative degradation of energy molecules like glucose.  
[during cellular respiration]
- Growth Pattern: Only at the Top of the medium.

#### → Microaerophiles: → نمو عند $PH$ المنخفضة قليلاً (أوسط الأوكسجين)

- Need small/limited amounts of Oxygen ( $O_2$ ).
- Excess  $O_2$  inhibits enzyme activity and causes death.
- Growth Pattern: Just below the surface of the medium.

#### → Facultative Anaerobes :

- Can grow with or without Oxygen.
- Prefer aerobic respiration when  $O_2$  is available.
- In low oxygen or no  $O_2$ , use anaerobic respiration with  $NO_3^-$  or  $SO_4^{2-}$  (Inorganic source of oxygen) as final electron acceptors.
- Growth Pattern: Throughout the medium, denser at the Top.

### → Aerotolerant Anaerobes:

- Don't use  $O_2$  but can tolerate its presence
- Produce Catalase and/or Superoxide dismutase, so  $O_2$  doesn't harm them.
- Metabolism: Fermentative (energy by fermentation).
- Growth Pattern: Evenly throughout the medium.

### → Obligate Anaerobic:

- Grow only in the absence of  $O_2$ .
- Use other molecules ( $NO_3^-$ ,  $SO_4^{2-}$ ) as final electron acceptors. (Use inorganic source of oxygen).
- $O_2$  forms toxic products like Superoxide ( $O_2^-$ )
- These organisms lack the enzymes Superoxide dismutase and Catalase.

#### • Detoxification Reactions:



- Without these enzymes, even small amounts of  $O_2$  are lethal.
  - Growth Pattern: At the bottom of the medium.
- 



## Section 3

### "Identification of Enteric Bacteria by IMViC test"

Enteric Bacteria: These are bacteria that live in the intestinal tract and belong to family Enterobacteriaceae.

→ They are classified according to their ability to ferment lactose into:

→ Lactose Fermenters.

- These bacteria can ferment lactose producing acid.
- So their colonies appear Pink or Red on MacConkey Agar.
- Example: E. Coli - Klebsiella Pneumoniae - Enterobacter.

→ Non-lactose Fermenters.

- These bacteria cannot ferment lactose, so their colonies appear colorless or pale on MacConkey agar.
- Example: Salmonella SPP. - Shigella SPP. - Proteus SPP.

IMViC test A group of biochemical tests used to differentiate enteric bacteria, especially lactose fermenters such as E. Coli - Klebsiella - Enterobacter.

\* IMViC stands for:

I → Indole test.

M → Methyl Red test.

V → Voges-Proskauer test.

C → Citrate Utilization test.

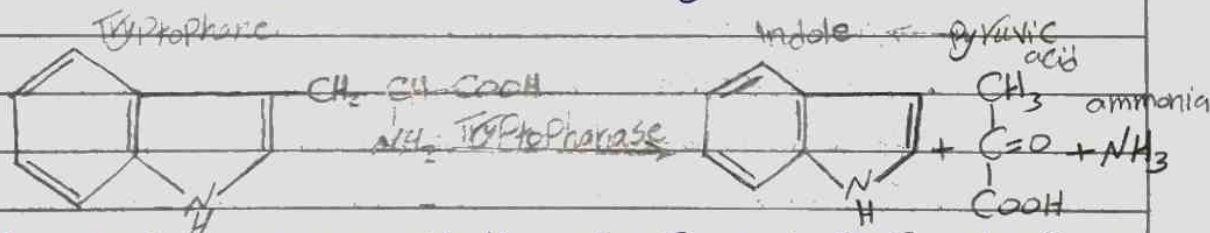
# Indol production test

- Aim:**
- 1- To differentiate between enteric bacteria (lactose formation) *E. coli* - *K. pneumoniae* or *E. aerogenes*.
  - 2- Ability can utilize tryptophan

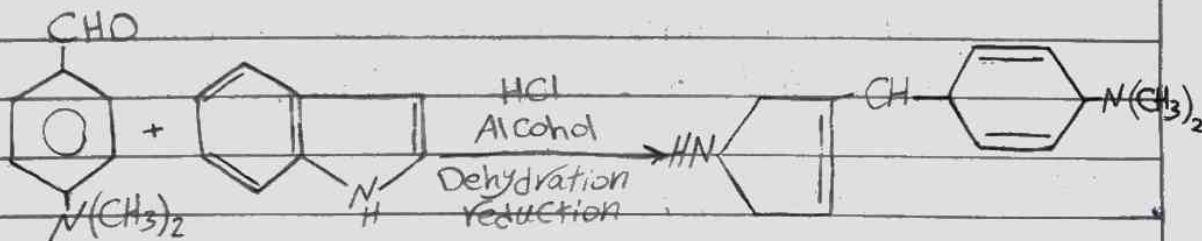
**Reagent** Kovac's reagent  
(p-dimethylaminobenzaldehyde + Butanol + HCl).

**Media** → SIM agar contains tryptophan.

**Principle** → Bacteria with the enzyme tryptophanase break down tryptophan into indole, pyruvic acid, and ammonia

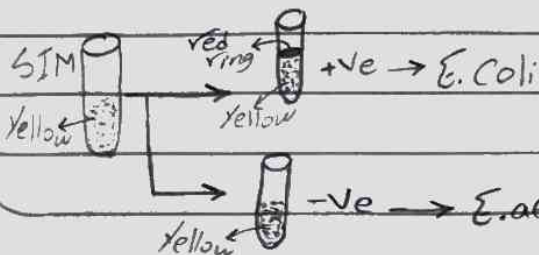


**I** → Indole reacts with Kovac's reagent to form a charged-red color on the surface (red ring).



**Result**

- Red ring on the surface → *E. coli*
- No color change → *E. aerogenes* or *Klebsiella*



(at 37°C → 24 hour)



# Methyle Red Test

Aim → To differentiate between *E. coli* and *K. pneumoniae* or *E. aerogenes*.

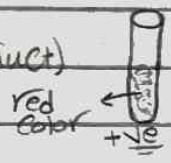
Reagent → MR (Methyl Red indicator)

Media → MR. VP. broth media.

Principle → *E. coli* → Performs mixed acid fermentation, keeps  $\text{pH} \leq 4.4$  → Red color (positive).  
*E. aerogenes* → Converts acids to neutral compounds →  $\text{pH} \approx 6$  → Yellow color (negative).

glucose oxidation → acids + gas ( $\text{CO}_2$ )

→ acid + MR → Remain constant → acid (end product)  
 $\text{pH} = 4$  → *E. coli*



→ Change → neutral compounds  
 Ex: Acetyl methyl carbide →  $\text{pH} = 6$   
 → *E. aerogenes* or *K. pneumoniae*.



# Vogec's-Proskauer Test.

**Aim**

To differentiate between *E. Coli* and (*K. pneumoniae* or *E. aerogenes*).

**Reagent**

Barrett's reagent. is consists of alpha-naphthol and 40% KOH.

**Media**

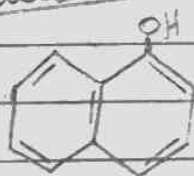
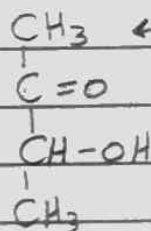
MR.VP broth media.

**Principle**

Acetoin (acetyl methyl carbonyl) is oxidized to diacetyl, which reacts with guanidine groups in Pepton (in presence of  $\alpha$ -naphthol and 40% KOH) forming a pink/red complex. (pH=8)

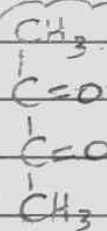
Glucose Fermentation  $\rightarrow$  acid + gas convert non-acidic compound

acetyl methyl carbonyl (Neutral compound)

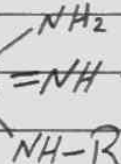


$\alpha$ -naphthol

40% KOH  
Oxidation



Diacetyl



guanidine group of Peptone

Pink color

Red color

**Result**

Positive: Rose-Pink/red color  $\rightarrow$  Acetoin Produced  
 $\therefore$  *Enterobacter aerogenes* - *K. pneumoniae*.

Negative: No color change  $\rightarrow$  *E. Coli*

MR.VP



red color  
+ve

*E. aerogenes* or *K. pneumoniae*



yellow color  
-ve

*E. Coli*



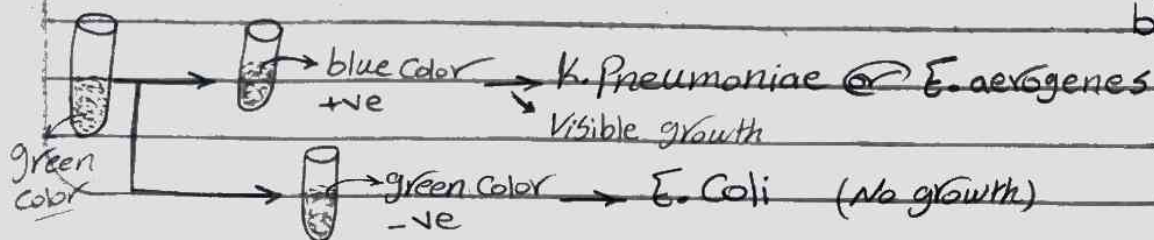
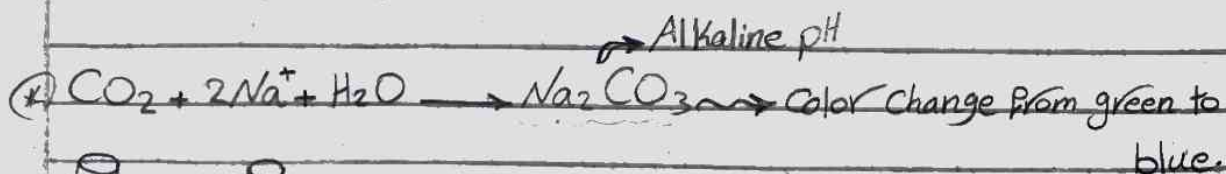
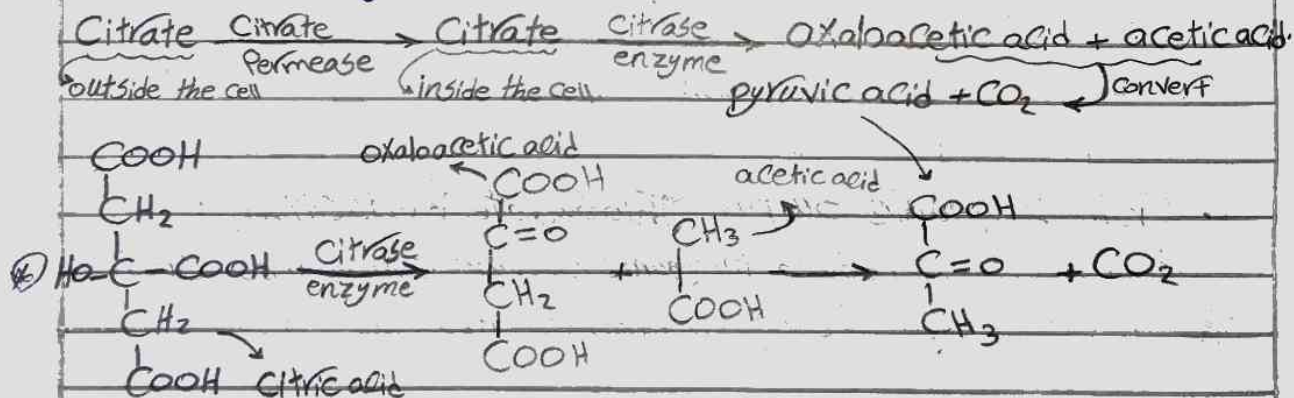
## Citrate Utilization test

**Aim** → To determine the ability of some bacteria to use Citrate as a sole carbon source.

**Reagent** → Bromothymol blue. (within the medium)

**Media** → Simmons' Citrate agar slants.

**Principle** → Bacteria with Citrate permease transport Citrate into the cell, where Citrase enzyme converts it into Oxaloacetic acid and acetic acid. These are converted to Pyruvic acid and  $\text{CO}_2$ .  $\text{CO}_2$  combines with sodium and water to form  $\text{Na}_2\text{CO}_3$  an alkaline product that turns bromothymol blue from green to blue.



## Hydrogen Sulfide test

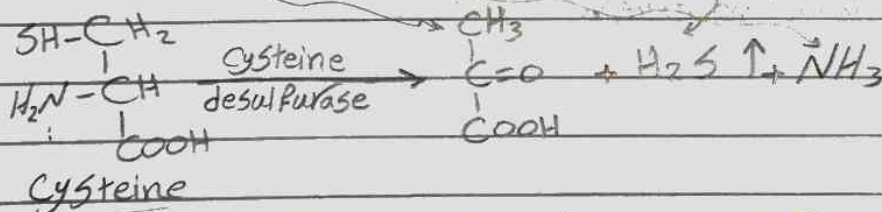
**Aim** → To differentiate between bacteria according to Production  $H_2S$  ex. Salmonella (+ve) - Shigella (-ve)

**Reagent** → Ferrrous ammonium sulfate ( $Fe(NH_4)_2SO_4$ )  
 $Fe(NH_4)_2SO_4 + H_2S \rightarrow FeS$  Black ppt.

**Media** → SIM media

**Principle** → Certain microorganisms (bacteria) can generate  $H_2S$  gas through two main pathways:

① From Organic source of Sulfur (Amino Acid: Cysteine):  
Some bacteria contain the enzyme cysteine desulfurase which breaks down cysteine (an amino acid in peptides) to produce pyruvic acid, ammonia and hydrogen sulfide gas ( $H_2S$ ).



② From Inorganic Sulfur (Thiosulfate Reduction):

Other bacteria reduce inorganic sulfur compounds such as ~~the~~ Thiosulfate ( $Na_2S_2O_3$ ) to produce  $H_2S$  gas.

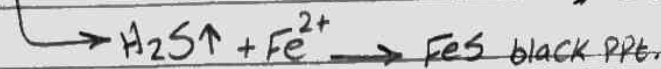


The released gas reacts with ferrous ions ( $Fe^{2+}$ ) in the medium to form black precipitate (Ferrous sulfide  $FeS$ )

Thus, black color formation indicates positive  $H_2S$  production.



∴ -ve → Shigella



∴ +ve → Salmonella.





# Catalase Test

Aim → To differentiate between bacteria according to production of Catalase enzyme.

Reagent → 3% hydrogen peroxide ( $H_2O_2$ )

Media → Trypticase (Tryptose) soy agar slant.

Principle → During aerobic respiration, some microorganisms produce hydrogen peroxide ( $H_2O_2$ ) and superoxide radicals ( $O_2^-$ ), both toxic to cell.

→ Organisms that produce catalase or peroxidase can break down  $H_2O_2$  into water and oxygen gas, preventing toxicity.



→ Organisms that lack catalase, peroxidase, or superoxide dismutase cannot degrade  $H_2O_2$  and thus are harmed by oxygen.

→ When 3% hydrogen peroxide is added to a bacterial colony, exposure

→ Bubble formation ( $O_2$  release) = Positive test  
(Catalase present)

→ No bubble = Negative test (no catalase enzyme).

Section 5Bacterial Growth CurvePurposes

→ To study the growth dynamics of bacterial cultures.

→ To plot a bacterial growth curve and determine the generation time (GT).

① → Bacteria, when inoculated into a sterile broth medium and incubated under optimal Temp., pH, and Oxygen conditions, multiply rapidly.

② → The increase in cell number with time can be represented as a bacterial growth curve, which helps visualize the growth cycle and calculate the generation time.

Generation Time → The time required for a microbial population to double.

③ → The growth curve is divided into four distinct phases:

① Lag Phase → Cells adapt to the new environment; no immediate cell division occurs.  
→ Metabolic activity is high as enzymes and molecules are synthesized.

② Log Phase  
"Exponential Phase" → Cells divide at a constant and rapid rate; Population doubles regularly.  
→ The generation time (GT) is calculated during this phase.

③ Stationary Phase → Growth rate = Death rate.  
→ Nutrients become limited, and waste products accumulate.



#### ④ Death (Decline) Phase

Nutrients are exhausted; toxic materials increase.

The number of viable cells decrease.

#### \* → Determine the generation time (GT).

# Indirect Method (Optical Density Method).

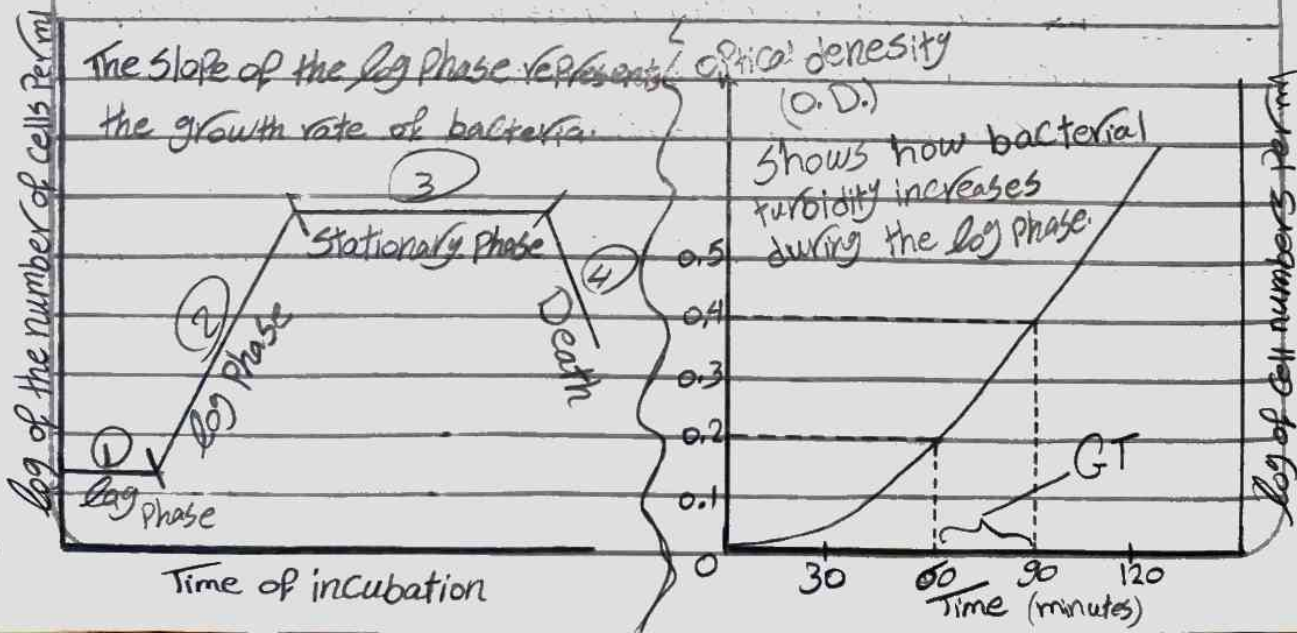
\* Measure the optical density (O.D.) to monitor turbidity over time. \* Select two O.D. points representing a doubling (e.g. 0.2 & 0.4), then determine their corresponding times ( $t_1$  &  $t_2$ ):  $GT = t_2 - t_1$

# Direct Method (Cell Count Method).

\* This uses the actual cell numbers (log scale) during the log phase:

$$GT = \frac{T \log 2}{\log b - \log B} \quad \text{where:}$$

- GT = Generation time.
- B = Cell number at first point of log phase.
- b = ~ ~ ~ second ~ ~ ~
- T = Time between B & b (hours or minutes).



## Extra Cellular Enzymes

### # Starch hydrolysis test :

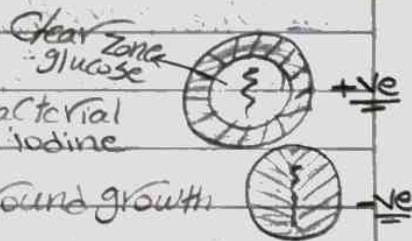
**Aim:** To detect the ability of microorganisms to produce amylase, which breaks down starch into maltose & glucose.

**Principle:** Starch (a polysaccharide) is hydrolyzed by Amylase into smaller molecules: dextrins  $\rightarrow$  maltose  $\rightarrow$  glucose

**Medium:** Starch agar (nutrient agar + starch)

**Reagent "indicator" :** Iodine Solution

Starch  $\xrightarrow{\text{amylase enzyme}}$   $\begin{cases} \text{Clear Zone around bacterial growth after adding iodine} \\ \text{Blue black color around growth} \end{cases}$



### # Lipid hydrolysis "Tributyrin test"

**Aim:** To determine the ability of bacterial to produce lipase, which hydrolyzes triglycerides into glycerol and fatty acids.

**Principle:**  $\rightarrow$  Lipase breaks ester bonds in triglycerides by adding water (hydrolysis).  $\rightarrow$  Products are used in ATP Production and other biosynthetic Pathways.

**Medium:** Tributyrin agar (nutrient agar + triglyceride tributyrin)

Triglyceride  $\xrightarrow[\text{enzyme}]{\text{Lipase}}$   $\begin{cases} \text{Clear Zone around growth} \\ \text{glycerol} \rightarrow \text{fatty acids} \\ \text{opaque medium remains} \end{cases}$





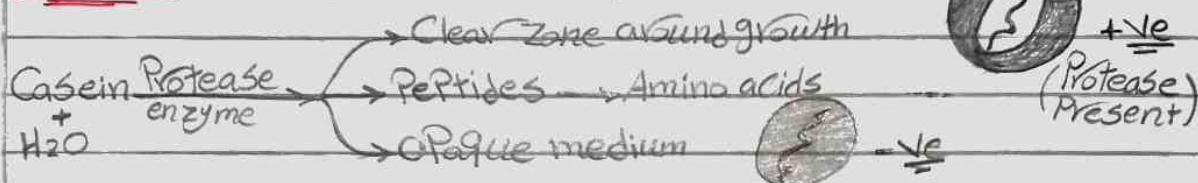
## Casein hydrolysis test

HgCl<sub>2</sub>

**Aim:** To test the ability of microorganisms to secrete proteases that degrade casein, the main protein in milk.

**Principle:** → Proteases hydrolyze peptide bonds ( $-CO-NH-$ ) in casein into soluble amino acids.  
→ The opaque milk medium becomes clear when casein is broken <sup>down</sup>.

**Medium:** Milk agar (nutrient agar + milk).



## Gelatin hydrolysis Test

**Aim:** To determine whether bacteria produce gelatinase, an enzyme that hydrolyzes gelatin into amino acids.

**Principle:** → Gelatin (a protein from collagen) solidifies below 25°C but liquefies when hydrolyzed.  
→ Gelatinase breaks peptide bonds, converting solid gelatin into <sup>liquid</sup>.

**Medium:** Nutrient gelatin deep tubes (12% gelatin).

